

Statistical Modeling and Naïve Bayes

Data Mining
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Statistical Modeling

- We looked at the simplicity of 1R where we only looked at one attribute
- Statistical Modeling uses all the attributes
- Two important assumptions:
 - All attributes are equally important
 - All attributes are statistically independent
(Knowing the value of one attribute does not tell us anything about the value of another – this is not always correct!)

Basics of Naïve Bayes

- Uses probability theory to find the best classification
- Uses data from trials as the basis for the results
- Assumes, of course, that the trial data is truly representative of real life

What is real life?

- Suppose we have data gathered during the last N days
- Data tracks whether a train is cancelled, on-time, early or late
- Probabilities of each are used to project unknown scenarios which have not yet been seen (otherwise we would have the actual data)

Flaws in the trial data

- What happens if all the trial (sample) data was gathered in Rochester during the month of January?
- How could this skew the results?
- Is the data a valid representation of the train behavior?

Probability theory

- The probability of an event is a number from 0 to 1, inclusive
- Probability of 0 means it won't occur
- Probability of 1 means it is certain to occur
- A probability of .7 implies that based on our trial data, the event has a 70% chance of occurring

Probability theory (2)

- If you have exactly N mutually exclusive and exhaustive events, the probabilities of all N sum to 1

The train example

- With events:
 - E1 – train cancelled
 - E2 – train 10 minutes or more late
 - E3 – train less than 10 minutes late
 - E4 – train on-time or early
 - $P(E1) = 0.05$
 - $P(E2) = 0.10$
 - $P(E3) = 0.15$
 - $P(E4) = 0.70$
- These probabilities sum to 1 because only one may occur at any time

Day	Season	Wind	Rain	Class
weekday	spring	none	none	on-time
weekday	winter	none	slight	on-time
weekday	winter	none	slight	on-time
weekday	winter	high	heavy	late
Saturday	summer	normal	none	on-time
weekday	autumn	normal	none	very late
holiday	summer	high	slight	on-time
Sunday	summer	normal	none	on-time
weekday	winter	high	heavy	very late
weekday	summer	none	slight	on-time
Saturday	spring	high	heavy	cancelled
weekday	summer	high	slight	on-time
Saturday	winter	normal	none	late
weekday	summer	high	none	on-time
weekday	winter	normal	heavy	very late
Saturday	autumn	high	slight	on-time
weekday	autumn	none	heavy	on-time
holiday	spring	normal	slight	on-time
weekday	spring	normal	none	on-time
weekday	spring	normal	slight	on-time

Credit: Bramer, Principles of Data Mining, Springer, 2007

Given the train data

- What is the classification for a weekday train, winter, high wind and heavy rain?
- Easy, but flawed, method is to just pick the most common classification
- With this data, we would be correct 70% of the time (on-time is the most common; $P(\text{on-time}) = 0.70$). This is the *prior* probability (14/20)

Attribute data

- The additional attribute data (wind, rain, season) give us better info about the classification
- We can compute the *conditional* probability ***given that*** other conditions exist
- Example: What is the probability of the train being on-time given that the season is winter?
- Written as
 $P(\text{class} = \text{on-time} \mid \text{season} = \text{winter})$

Conditional Probability

- The probability of an event if we know that an attribute has a particular value (or several attributes have values)
- $P(\text{class} = \text{on-time} \mid \text{season} = \text{winter})$
- Also called the *posterior* probability – we calculate the probability after we have obtained the information that the season is winter
- (To contrast, the *prior* probability is estimated before any other info is available)

Naïve Bayes

- Naïve Bayes gives us a way to combine prior probabilities and conditional probabilities to calculate the probability of each classification (as outlined on the previous slide)
- *Naïve* implies that the probabilities are independent of the values of the other attributes – which may not always be valid

Naïve Bayes Classification

- Given a set of k mutually exclusive and exhaustive classifications c_1, c_2, c_k , which have prior probabilities $P(c_1), P(c_2), \dots P(c_n)$, with attributes $a_1, a_2, \dots a_n$ for given instance values $v_1, v_2, \dots v_n$: calculate using the product

$$P(c_i) \times P(a_1 = v_1 \mid c_i) \times P(a_2 = v_2 \mid c_i) \times \dots \times P(a_n = v_n \mid c_i)$$

- Multiply the prior probability * the conditional probabilities

To answer our previous question:

- What is the class for a weekday train, winter, high wind and heavy rain?
- Calculate:
 - $P(\text{class} = \text{on-time} \mid \text{day} = \text{weekday AND season} = \text{winter AND wind} = \text{high AND rain} = \text{heavy})$
 - $P(\text{class} = \text{late} \mid \text{day} = \text{weekday AND season} = \text{winter AND wind} = \text{high AND rain} = \text{heavy})$
 - $P(\text{class} = \text{very late} \mid \text{day} = \text{weekday AND season} = \text{winter AND wind} = \text{high AND rain} = \text{heavy})$
 - $P(\text{class} = \text{cancelled} \mid \text{day} = \text{weekday AND season} = \text{winter AND wind} = \text{high AND rain} = \text{heavy})$
- Pick the highest probability of the 4

Performance of Naïve Bayes

- Works well, even if the independence assumption is violated
 - Classification doesn't require accurate probability estimates as long as the maximum probability is assigned to the correct class
- But adding too many redundant attributes will cause problems (identical attributes)

Performance of Naïve Bayes

- Often works well or better with only a fraction of all attributes
- Must have categorical data. May need to convert continuous values

Zero Frequencies

- What should be done if the attribute data does not occur with every class value?
- Suppose data showed all High Humidity days == Yes. This would mean that other values would be zero – resulting in a zero probability
- Solution: Add a very small value to the count for every attribute. Ensures you will never have zero

Weights

- Some attributes may be more important or more likely to occur
- We can also associate a constant with each attribute
- Weights do not need to be equal but they must sum to one

In-Class Example

- Using the Weather Data, how would you classify a Sunny, Cool, High Humidity, Windy Day? Play or not?

Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Bayes Self-Check Example

- Use the data on the following slide
- What is the classification for age ≤ 30 , income = medium, student = yes and creditRating = fair?

Age	Income	Student	CreditRate	Purchase?
<=30	High	No	Fair	No
<=30	High	No	Excellent	No
31..40	High	No	Fair	Yes
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	Excellent	No
31..40	Low	Yes	Excellent	Yes
<=30	Medium	No	Fair	No
<=30	Low	Yes	Fair	Yes
>40	Medium	Yes	Fair	Yes
<=30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
>40	Medium	No	Excellent	No